

Lec 34 GAN Convergence and Nash equilibrium

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Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and the large language models. We are in the week five and in the previous session, we had seen the GAN objective function and then the loss function. And in this particular session, we will be seeing something called as a GAN convergence and a Nash equilibrium. So, let's get us started. So, if you look into this, all the real training samples are If your X is a real training samples, all these samples are drawn from a real data distribution P data of X . So, this is with respect to real samples. When you take in a generated samples, how is that you get the generated samples? Is it not the noise has to first sample the from this

[00:49] particular distribution and this Z has to be passed to an generator. Then generator is going to produce the samples. So, here we are seeing something called as a P data of X and then we have seen how the generated samples are produced. Now, once all the samples have been produced, that is a generated samples, is it not all those samples will follow the generated distribution? Let me make this point again. So, all the real samples are drawn from a real data distribution, that is P data of X .

[01:29] Now, Z is being sampled from the normal distribution. That Z is a passed to an a generator and generator is going to produce the generated samples. Now,

think of all the collection of such a generated samples, is it not that these generated samples belongs to some distribution? That distribution is a generated distribution. Right? The collection of all generated samples will follow the generated distribution. Therefore, P data of X is actually a real data distribution and the PG of X is the distribution of samples generated by the generator G 5.

[02:12] Now, if you see initially, you have still not started anything at the beginning of a training, the generator has not at all been trained properly because you have randomly assigned the values. When you have randomly assigned the values, 5 is whatever is the generator parameter 5 is not well a trained parameter. So, when it is not a trained one, whatever the samples that it is going to produce, the generated samples, these are the generated samples, whatever samples the generator is going to produce is it not very noisy or meaningless or unrealistic samples?

[02:54] So, at this point of time, is it not the Now, we have to talk with respect to these two. Whatever is the GAN convergence or Nash equilibrium, we talk with respect to these two terms. So, this is the real data distribution and the PG of X is the generated distribution. Now, when the parameters are not been trained during the initial phase of a training, definitely whatever the parameter whatever the parameters are not well trained parameters. As a resultant of this, the generated samples are very poor. So, can we say what will be the relation between this and this?

[03:33] Definitely, they are not at all equal. Generator produces very good images, they look very much similar to realistic real images. When they look real when they look similar to real, it means is it not that it is it looks similar to a this particular real data distribution, right? So, at the beginning, these two are not equal. So, generated distribution is a different from a real data distribution. These two are not equal. When these two are not equal, which means for example, let us take a human face data set. All real images which are drawn from this particular distribution really looks like a face images. And since here at the beginning, these two are not the same, whatever the generated samples

[04:28] which are generated from this particular from this particular step, definitely these looks very noisy one. That's what is being written here. For a face data set, for a real samples look like a human face. And the generated samples may look like noise or you know, disturbed images or unrealistic uh images. So, this is what Now, at this point of time, what is the role of a discriminator? How discriminator is a performing? When real images are passed, discriminator is giving the probability closer to one.

[05:06] When a generated images are passed to a discriminator, discriminator is giving closer to zero. So, that is what uh the discriminator is doing. When real images are passed, it is giving the values closer to one. When generated samples are passed, it is giving the value closer to zero. Which means at this point of time, at the initial phase, is it not a discriminator is performing very well while generator is performing

very poorly, right?

[05:35] Discriminator performs well, but generator performs poorly. Okay. So, with this, what we are doing? We are now progressing. We are you know, increasing the training. As the training progresses, what exactly happens? Is it not a discriminator is initially discriminator has given one closer to one for real images and the closer to you know zero for generated images, which means the discriminator is performing well. So, as the training progresses, discriminator provides the gradients back to a generator, right?

[06:11] So, as the training progresses, the discriminator provides the gradients that are back propagated through discriminator to the generator, right? So, through although discriminator is there, but those parameters are not being updated. So, which means once a discriminator makes some prediction, you calculate the loss and then that feedback is given to a generator. Now, when the feedback is given to a generator, what generator will do?

[06:40] Generator will update its parameters. What parameters of ϕ here? So, generator updates its parameters of ϕ . So, when generator updates its parameters, now what is going to improve? The generated samples are going to improve. So, over a period of time, what exactly is happen what exactly happens? We are talking with respect to those two relation. So, over a period of time, is it not that this generated distribution moves closer to a real data distribution?

[07:14] As the generated samples improves, the generated distribution move closer to a real data distribution. Okay. If it moves closer to real data distribution, what is the other meaning of this? Is it not now whatever the samples that the generated has generated, is it not very much closely matching to a real images? So, at a sample level, it means the generated samples are very much realistic to our closer to real images.

[07:46] Therefore, it becomes harder for a discriminator to say whether what is a real and what is generated one. I hope up till this it is a clear to you. Now, uh as the generator improves, what what exactly improves? Is it not this value gets increasing? So, when this value gets increases, initially maybe this value is 0.1. But as later the value is 0.4. Further, when you improve the value will become 0.5. So, which means the generated images are becoming more and more difficult for the discriminator to classify that it is a fake images. So, let me explain you once again. This is simple concept.

[08:37] Initially, the real images are taken from the distribution P data of X . This is the samples which are taken from the real distribution. When the samples are taken from a real distribution and when these samples are passed to an discriminator and a discriminator aim is to give the probability closer to one. Now, you have taken and generated samples. Generated samples are these. These generated samples are

given to a discriminator.

[09:09] Now, initially discriminator performance will be very good because generator has not performed very good images. Therefore, we can say that the generated samples are very much away. Initially, the generated samples are very much away from the real data distribution. But as the training goes on, the discriminator is giving the feedback to a generator. So, generator will start improving. When generator starts improving, it also improves the generated samples. When generated samples gets improved, is it not that it's a distribution move closer to a real data distribution?

[09:54] When it moves closer to real data distribution, it means that the samples what the generator has generated is now closely matching to real data. When it matches closer to real data, is it not the discriminator finds it hard to distinguish? Right? This is what we have discussed. So, at an ideal equilibrium, you know, this is the case. At an ideal equilibrium, PG of X is equivalent to P data of X , which means that the generated distribution is closely matching or is matching to a real data distribution. Or come sample come closer to a real data distribution. If this is the case, can we say that discriminator has now reached to a state where it can no longer distinguish between real samples

[10:53] and generated samples? Can we tell that? Now, at one or the other point of time, this generator has will reach to a stage where it starts generating the images closer to real images. When it starts generating images closer to real images, discriminator now finds it hard. Discriminator now confuses that whether it's a real image or it's a generated image. The state where Think about the discriminator now. The state where discriminator cannot identify We have now reached to this particular stage. What do you mean by this? The real data distribution and generated data distributions are identical. When the at this point of time, whatever the images that you pass as an input to a discriminator, discriminator will give us this

[11:45] probability. So, which means for a real image, it gives this probability. For an generated images also, it gives this probability. So, what exactly that 0.5 indicates discriminator is not poorly performing. The meaning of this is generator has now reached to a stage where it is it started producing the images which are closer to a real-time or a real data set. Okay. So, therefore, what exactly is GAN convergence, GAN convergence? GAN convergence, it is a state in which the generated data distribution uh matches to a real data distribution.

[12:32] Generated This is a generated data distribution, and this is a real data distribution. This generated data distribution matches to a real data distribution, and discriminator can no longer distinguish between real images from generated images. So, this is called an GAN convergence. GAN convergence is a state in which the generated data distribution matches the real data distribution, which means discrim-

inator can no longer distinguish between real images and generated samples. This state we call it as an GAN convergence, or we also call this as an Nash equilibrium. What is a Nash equilibrium here? There are two players, generator and a discriminator, and at equilibrium, this generated data

[13:31] distribution is matching to an real data distribution. Therefore, any image that you pass as an input to a discriminator, discriminator is giving this probability 0.5. Which means discriminator can no longer improve. Why it can no longer improve? Because real and generated samples are now following the same distribution. At the same time, generator also cannot further improve. Why it has a started producing the images which are matching to an real data distribution.

[14:05] So, therefore, what exactly is a Nash equilibrium? Nash equilibrium is a state in which neither generator nor discriminator are improving. Right? You cannot further improve. Neither a generator nor a discriminator can obtain a better outcome by changing its own parameters alone, while the parameters of other network are kept unchanged. So, which means you cannot change you cannot improve only discriminator alone. You cannot improve only generator alone. You cannot improve it by keeping other network constant, fixed. Why? Because both have reached to a stage where you cannot further improve. This is called a Nash equilibrium. In one statement, if you have to tell about GAN convergence and a Nash equilibrium is

[15:02] generator has a started generating an images which are as close as the real images. Which means generated data distribution is now matching to real data distribution. If that is the case, discriminator can no longer distinguish between real images and the images. This is a called a convergence and a Nash equilibrium. So, with this we have come to an end of this particular session where in this we have seen the convergence and a Nash equilibrium. And the whole concept here is this. One, mathematically if you have to tell, generated samples are matching with the real sample distribution. Because of this, whatever input here X is not real now. Whatever input you pass it to two and discriminator discriminator is now

[15:51] confused whether it's a real image or whether it's an generated image. That state neither discriminator can further improve nor generator can further improve. This is called a Nash equilibrium. So, in the upcoming session we will see a concept called a DCGAN, that is a deep convolutional GAN. So, if you have to see this particular when you hear this a term DCGAN, what exactly we will understand from this is whatever we have seen so far is the basic GAN. So, which means a basic GAN can be used to for an images for a data such as images or you can even pass in a tabulated data. That is the data in the form of a CSV file.

[16:35] So, whereas when you talk about a DCGAN, DCGAN is specially designed for an images. So, this architecture you will see in detail in the upcoming session. Thank you. »